

Project ParaBank: Banking Automation Testing Framework

Selected Website

ParaBank Banking Application
<https://parabank.parasoft.com/parabank/index.htm>

Project Objective

To build a scalable and enterprise-grade automation testing framework for the ParaBank banking application using Playwright and JavaScript capable of validating real-world banking workflows such as authentication, account management, fund transfer, bill payment, transaction handling, session management, loan processing, and API validations.

Technology Stack

Technology	Purpose
Node.js	Runtime Environment
Playwright	Automation Framework
JavaScript	Programming Language
Page Object Model (POM)	Framework Design
HTML Reports / Allure	Test Reporting
GitHub	Version Control
Jenkins	CI/CD Integration

Planned Modules & Scope

The automation suite contains exactly 8 core banking service modules, with each module executing 15 distinct automated test cases totaling 120 scenarios.

1. Login & Authentication Module
2. User Registration Module
3. Open New Account Module
4. Accounts Overview Module
5. Transfer Funds Module

6. Bill Pay Module
7. Find Transactions Module
8. Request Loan + API Testing Module

Planned Features & Technical Implementations

Service Module	Technical Implementation & Focus
Service 1: Authentication	Login validation, session persistence, secure credential handling, logout verification
Service 2: Registration	Dynamic form validation, duplicate user validation, mandatory field verification
Service 3: Open New Account	Account creation workflows, dropdown validations, account synchronization
Service 4: Accounts Overview	Balance verification, account detail extraction, dynamic data validations
Service 5: Transfer Funds	Transaction workflows, balance updates, concurrent transfer handling
Service 6: Bill Pay	Payee management, bill payment validation, payment confirmation workflows
Service 7: Find Transactions	Search filters, transaction sorting, date-range validations
Service 8: Loan & API Testing	REST API validation, session token handling, request/response verification

Framework Structure

```
Playwright-ParaBank/  
|  
├── tests/  
|   ├── authentication/  
|   ├── registration/  
|   ├── openAccount/  
|   └── accountOverview/
```

```
|
| ├── transferFunds/
| ├── billPay/
| ├── transactions/
| └── loanAndApi/
|
| ├── pages/
| ├── utils/
| ├── fixtures/
| ├── reports/
| ├── screenshots/
|
| ├── playwright.config.js
| ├── package.json
| └── README.md
```

Expected Outcomes

- 120 Fully Automated Test Cases
- Reusable Playwright Automation Framework
- Cross-browser Testing Support
- HTML Reporting and Screenshot Capture
- API + UI Integration Testing